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Electronic device and manufacturing method of the same

Abstract

According to one embodiment, an electronic device includes a lower electrode, a second insulating layer including an opening overlapping the lower electrode, an organic layer covering the lower electrode at the opening, an upper electrode covering the organic layer, and a barrier wall on the second insulating layer. The barrier wall includes a first layer formed of a metal material and including a first side surface which is in contact with the upper electrode, a second layer disposed above the first layer and extending from the first side surface toward the opening, and a protective layer formed of a material different from a material of the organic layer and covering at least an upper surface of the second layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2021-092928, filed Jun. 2, 2021, the entire contents of which are incorporated herein by reference.

FIELD

(2) Embodiments described herein relate generally to an electronic device and a manufacturing method of the same.

BACKGROUND

(3) Recently, display devices with organic light-emitting diodes (OLEDs) applied thereto as display elements have been put into practical use. Such a display element comprises an organic layer between a first electrode and a second electrode. For example, the organic layer is formed by vacuum deposition and the second electrode is formed by sputtering.

(4) For example, in deposition using a mask, a fine mask including openings corresponding to respective pixels is applied. However, formation accuracy of a thin film formed by deposition may be deteriorated due to processing accuracy of the fine mask, deformation of the opening shape, and the like.

(5) Thus, a technique of dividing the organic layer and the second electrode using a pixel division structure is known.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a view showing a configuration example of an electronic device **100** according to embodiments.
- (2) FIG. 2 is a plan view showing an example of the pixel PX shown in FIG. 1.
- (3) FIG. 3 is a view showing an example of the configuration of the display element **20**.
- (4) FIG. 4 is a cross-sectional view showing the electronic device **100** as viewed along line A-B shown in FIG. 2.
- (5) FIG. 5 is a view illustrating a method of manufacturing the electronic device **100**.
- (6) FIG. 6 is a view illustrating a method of manufacturing the electronic device **100**.
- (7) FIG. 7 is an enlarged cross-sectional view showing another configuration example of the barrier wall **30**.
- (8) FIG. 8 is an enlarged cross-sectional view showing the other configuration example of the barrier wall **30**.

DETAILED DESCRIPTION

- (9) In general, according to one embodiment, an electronic device includes a substrate, a first insulating layer disposed on the substrate, a lower electrode disposed on the first insulating layer, a second insulating layer disposed on the first insulating layer and including an opening overlapping the lower electrode, an organic layer covering the lower electrode at the opening, an upper electrode covering the organic layer, and a barrier wall disposed on the second insulating layer. The barrier wall includes a first layer formed of a metal material and including a first side surface which is in contact with the upper electrode, a second layer disposed above the first layer and extending from the first side surface toward the opening, and a protective layer formed of a material different from a material of the organic layer and covering at least an upper surface of the second layer.
- (10) According to another embodiment, a method of manufacturing an electronic device, includes forming a lower electrode on a first insulating layer, forming a second insulating layer covering the first insulating layer and the lower electrode, forming a barrier wall on the second insulating layer, and forming an opening to expose the lower electrode in the second insulating layer. The barrier wall includes a first layer formed of a metal material and including a first side surface, a second layer disposed above the first layer and extending from the first side surface toward the opening, and a protective layer covering at least an upper surface of the second layer. The opening is formed by dry etching the second insulating layer using the barrier wall as a mask.
- (11) Embodiments will be described hereinafter with reference to the accompanying drawings.
- (12) The disclosure is merely an example, and proper changes in keeping with the spirit of the invention, which are easily conceivable by a person of ordinary skill in the art, come within the scope of the invention as a matter of course. In addition, in some cases, in order to make the description clearer, the widths, thicknesses, shapes and the like, of the respective parts are illustrated schematically in the drawings, rather than as an accurate representation of what is implemented. However, such schematic illustration is merely exemplary, and in no way restricts the interpretation of the invention. In addition, in the specification and drawings, structural elements which function in the same or a similar manner to those described in connection with preceding drawings are denoted by like reference numbers, detailed description thereof being omitted unless necessary.
- (13) In the drawings, an X-axis, a Y-axis and a Z-axis orthogonal to each other are described in the drawings to facilitate understanding as needed. A direction along the X-axis is referred to as an X-direction or a first direction, a direction along the Y-axis is referred to as a Y-direction or a second direction, and a direction along the Z-axis is referred to as a Z-direction or a third direction. A plane defined by the X-axis and the Y-axis is referred to as an X-Y plane. Viewing the X-Y plane is

referred to as planar view.

(14) The electronic device **100** according to the embodiments is, for example, a display device comprising a display element. The display element is, for example, an organic light emitting diode (OLED) comprising an organic light emitting layer. The display element can be applied as a light emitting element of an illumination device, and the electronic device **100** may be an illumination device comprising a light emitting element.

(15) In addition, the electronic device **100** of the embodiments may also be a sensor device comprising a sensor element. The sensor element is, for example, an organic photodiode (OPD) comprising an organic photoelectric conversion layer.

(16) FIG. **1** is a view showing a configuration example of the electronic device **100** according to the embodiments. The electronic device **100** comprises a display area DA where images are displayed, on an insulating base material **10**. The base material **10** is an insulating substrate, and may be a glass substrate or a flexible resin film.

(17) The display area DA comprises a plurality of pixels PX arrayed in a matrix in the first direction X and the second direction Y. Each of the pixels PX comprises a plurality of sub-pixels SP1, SP2, and SP3. For example, the pixel PX comprises a red sub-pixel SP1, a green sub-pixel SP2, and a blue sub-pixel SP3. In addition to the sub-pixels of the above three colors, the pixel PX may comprise four or more sub-pixels including a sub-pixel of the other color such as white. Alternatively, combination of the colors of the sub-pixels included in the pixel PX may be different from the combination of red, green, and blue.

(18) A configuration example of one sub-pixel SP included in the pixel PX will be described in brief.

(19) The sub-pixel SP comprises a pixel circuit **1** and a display element **20** driven by the pixel circuit **1**. The pixel circuit **1** comprises a pixel switch **2**, a drive transistor **3**, and a capacitor **4**. The pixel switch **2** and the drive transistor **3** are, for example, switch elements constituted by thin-film transistors.

(20) In the pixel switch **2**, a gate electrode is connected to a scanning line GL, a source electrode is connected to a signal line SL, and a drain electrode is connected to one of electrodes constituting the capacitor **4** and a gate electrode of the drive transistor **3**. In the drive transistor **3**, a source electrode is connected to the other electrode constituting the capacitor **4** and a power line PL, and a drain electrode is connected to an anode of the display element **20**. A cathode of the display element **20** is connected to a feed line FL inside or outside the display area DA. The configuration of the pixel circuit **1** is not limited to the example shown in the figure.

(21) The display element **20** is an organic light emitting diode (OLED) which is a light emitting element. For example, the sub-pixel SP1 comprises a display element that emits light corresponding to a red wavelength, the sub-pixel SP2 comprises a display element that emits light corresponding to a green wavelength, and the sub-pixel SP3 comprises a display element that emits light corresponding to a blue wavelength. The pixel PX can realize multicolor display by comprising a plurality of sub-pixels SP1, SP2, and SP3 of different display colors.

(22) However, the pixel PX may also be configured such that the display element **20** of each of the sub-pixels SP1, SP2, and SP3 emits light of the same color. Monochromatic display can be thereby realized.

(23) In addition, when the display element **20** of each of the sub-pixels SP1, SP2, and SP3 is configured to emit white light, a color filter opposed to the display element **20** may be disposed. For example, the sub-pixel SP1 may comprise a red color filter opposed to the display element **20**, the sub-pixel SP2 may comprise a green color filter opposed to the display element **20**, and the sub-pixel SP3 may comprise a blue color filter opposed to the display element **20**. Multicolor display can be thereby realized.

(24) Alternatively, when the display element **20** of each of the sub-pixels SP1, SP2, and SP3 is configured to emit ultraviolet light, multicolor display can be realized by disposing a light

conversion layer opposed to the display element **20**.

(25) FIG. **2** is a plan view showing an example of the pixel PX shown in FIG. **1**.

(26) The sub-pixels SP**1**, SP**2**, and SP**3** which constitute one pixel PX are formed in an approximately rectangular shape extending in the second direction Y and are arranged in the first direction X. The outline of each of the sub-pixels corresponds to an outline of a light emitting area EA in the display element **20**, but is shown in a simplified form and does not necessarily reflect the actual shape. It is assumed here that the light emitting area EA is formed in a rectangular shape having shorter sides extending in the first direction X and longer sides extending in the second direction Y. An insulating layer **12**, which will be described in detail later, is formed in a grating shape extending in the first direction X and the second direction Y, in planar view, to surround each of the sub-pixels SP**1**, SP**2**, and SP**3** or the display element **20** of each of the sub-pixels. Such an insulating layer **12** may be referred to as a rib, a barrier wall, a bank, or the like. The light emitting area EA is formed in the opening OP**1** of the insulating layer **12**. A plurality of openings OP**1** are arrayed in a matrix in the first direction X and the second direction Y.

(27) A barrier wall **30**, which will be described in detail later, is formed in a grating shape extending in the first direction X and the second direction Y, in planar view, and is disposed on the insulating layer **12**. Each of the sub-pixels SP**1**, SP**2**, and SP**3** is surrounded by the barrier wall **30**.

(28) The shape of the barrier wall **30** is not limited to the example shown in FIG. **2**, but may be the other shape such as a stripe shape. In addition, the layout of the sub-pixels SP or the openings OP**1** is not limited to the example shown in FIG. **2** either.

(29) FIG. **3** is a view showing an example of the configuration of the display element **20**.

(30) The display element **20** comprise a lower electrode (first electrode) E**1**, an organic layer OR, and an upper electrode (second electrode) E**2**. The organic layer OR is disposed on the lower electrode E**1**, and the upper electrode E**2** is disposed on the organic layer OR. The organic layer OR includes a carrier adjustment layer CA**1**, a light emitting layer EL, and a carrier adjustment layer CA**2**. The carrier adjustment layer CA**1** is located between the lower electrode E**1** and the light emitting layer EL, and the carrier adjustment layer CA**2** is located between the light emitting layer EL and the upper electrode E**2**. The carrier adjustment layers CA**1** and CA**2** include a plurality of functional layers.

(31) An example in which the lower electrode E**1** corresponds to an anode and the upper electrode E**2** corresponds to a cathode will be described.

(32) The carrier adjustment layer CA**1** includes a hole injection layer F**11**, a hole transport layer F**12**, an electron blocking layer F**13**, and the like, as functional layers. The hole injection layer F**11** is disposed on the lower electrode E**1**, the hole transport layer F**12** is disposed on the hole injection layer F**11**, the electron blocking layer F**13** is disposed on the hole transport layer F**12**, and the light emitting layer EL is disposed on the electron blocking layer F**13**.

(33) The carrier adjustment layer CA**2** includes a hole blocking layer F**21**, an electron transport layer F**22**, an electron injection layer F**23**, and the like, as functional layers. The hole blocking layer F**21** is disposed on the light emitting layer EL, the electron transport layer F**22** is disposed on the hole blocking layer F**21**, the electron injection layer F**23** is disposed on the electron transport layer F**22**, and the upper electrode E**2** is disposed on the electron injection layer F**23**.

(34) In addition to the functional layers described above, the carrier adjustment layers CA**1** and CA**2** may also include the other functional layers such as a carrier generation layer as needed or at least one of the above functional layers may be omitted in the carrier adjustment layers CA**1** and CA**2**.

(35) FIG. **4** is a cross-sectional view showing the electronic device **100** as viewed along line A-B shown in FIG. **2**.

(36) The display element **20** located in the center, of three display elements aligned in the first direction X shown in FIG. **2** as an example, will be focused.

(37) The electronic device **100** comprises a base material **10**, an insulating layer (first insulating

layer) **11**, an insulating layer (second insulating layer) **12**, a display element **20**, and a barrier wall **30**. The insulating layer **11** is disposed on the base material **10** and corresponds to an underlying layer of the display element **20**. The pixel circuit **1** shown in FIG. **1** is disposed on the base material **10** and is covered with the insulating layer **11**. The insulating layer **12** is disposed on the insulating layer **11**.

(38) The lower electrodes **E1** are disposed on the insulating layer **11**, in the display element **20**. A plurality of lower electrodes **E1** are the electrodes which are arranged at intervals in the first direction **X**, and which are disposed for the respective sub-pixels or the respective display elements. Each of the lower electrodes **E1** is electrically connected to the drive transistor **3** included in the pixel circuit **1** shown in FIG. **1** and may be referred to as pixel electrode, anode, or the like.

(39) The lower electrodes **E1** are transparent electrodes formed of, for example, a transparent conductive material such as indium tin oxide (ITO) or indium zinc oxide (IZO). The lower electrodes **E1** may be metal electrodes formed of a metal material such as silver or aluminum. Alternatively, the lower electrodes **E1** may be stacked layer bodies of transparent electrodes and metal electrodes. For example, the lower electrode **E1** may be constituted as a stacked layer body formed by stacking a transparent electrode, a metal electrode, and a transparent electrode, in this order, or may be constituted as a stacked layer body of four or more layers.

(40) The insulating layer **12** is disposed between the adjacent lower electrodes **E1**. In addition, the insulating layer **12** includes an opening **OP1** and an upper surface **U1**. The insulating layer **12** is, for example, an inorganic insulating layer formed of a silicon nitride or the like. The opening **OP1** is a through hole which is formed in an area overlapping the lower electrode **E1** and which penetrates the insulating layer **12** to the lower electrode **E1**. A peripheral part of the lower electrode **E1** is covered with the insulating layer **12**, and a central part of the lower electrode **E1** is exposed from the insulating layer **12** at the opening **OP1**.

(41) The organic layer **OR** is disposed in the opening **OP1** and covers the lower electrode **E1**. In the example shown in FIG. **4**, the organic layer **OR** is also disposed on a part of the upper surface **U1**. Such an organic layer **OR** is disposed for each sub-pixel or each display element, similarly to the lower electrode **E1**.

(42) The upper electrode **E2** is stacked on the organic layer **OR** and covers an entire body of the organic layer **OR** including the peripheral part of the organic layer **OR**. In addition, the upper electrode **E2** is in contact with the upper surface **U1** outside the organic layer **OR**.

(43) A part of the organic layer **OR**, which is located between the lower electrode **E1** and the upper electrode **E2**, without intervening in the insulating layer **12**, can form a light emitting area of the display element **20**.

(44) The upper electrode **E2** is an electrode disposed for each sub-pixel or each display element, and a plurality of upper electrodes **E2** are electrically connected to each other through the barrier wall **30** as described later. The upper electrode **E2** may be referred to as a common electrode, a counter-electrode, a cathode or the like.

(45) The upper electrode **E2** is, for example, a semi-transparent metal electrode formed of a metal material such as magnesium or silver. The upper electrode **E2** may be a transparent electrode formed of a transparent conductive material such as ITO or IZO. Alternatively, the upper electrode **E2** may be a stacked layer body of transparent electrodes and metal electrodes.

(46) The barrier wall **30** is located between the adjacent display elements **20** and disposed on the insulating layer **12**. The barrier wall **30** is separated from the organic layer **OR** and is in contact with the upper electrodes **E2**. A left barrier wall **30**, of two barrier walls **30** shown in FIG. **4**, we will be focused and described in more detail. The barrier wall **30** on the right side of the figure has the same structure as the barrier wall **30** on the left side.

(47) The barrier wall **30** includes a first layer **31**, a second layer **32** disposed on the first layer **31**, and a protective layer **PR** covering at least a part of the second layer **32**.

(48) The first layer **31** is, for example, a conductive layer formed of a metal material. The first layer

31 is located on the insulating layer **12** and is in contact with the upper surface **U1**. In addition, the first layer **31** is disposed between adjacent organic layers **OR** and between adjacent upper electrodes **E2**.

(49) The first layer **31** has a side surface (first side surface) **S11** which faces the opening **OP1** (or the display element **20**) on the left side of the figure, a side surface **S12** which faces the opening **OP1** (or the display element **20**) in the center of the figure, and an upper surface **U11** between the side surfaces **S11** and **S12**. Each of the side surfaces **S11** and **S12** is in contact with the upper electrode **E2**. The upper electrodes **E2** adjacent to each other with the barrier wall **30** sandwiched therebetween are thereby electrically connected to each other.

(50) The second layer **32** is formed of a material different from that of the first layer **31**, for example, an insulating material. For example, the second layer **32** is an inorganic insulating layer formed of a silicon nitride or the like, which is the same material as the insulating layer **12**. The second layer **32** is in contact with the upper surface **U11**. In addition, the second layer **32** extends from the side surface **S11** toward the opening **OP1** on the left side of the figure, and further extends from the side surface **S12** toward the opening **OP1** in the center of the figure.

(51) The second layer **32** has a side surface (second side surface) **S21** which faces the opening **OP1** on the left side of the figure, a side surface **S22** which faces the opening **OP1** in the center of the figure, and an upper surface **U21** between the side surfaces **S21** and **S22**. In addition, the second layer **32** has a bottom surface **B21** between the side surfaces **S21** and **S22**. A width of the bottom surface **B21** along the first direction **X** is larger than that of the upper surface **U21** along the first direction **X**.

(52) The protective layer **PR** is disposed to cover the upper surface **U21** of the second layer **32**. In the example illustrated, the side surfaces **S21** and **S22** are exposed from the protective layer **PR**. The protective layer **PR** is formed of a material different from the organic layer **OR**. The protective layer **PR** is, for example, a metal layer which is formed of a metal material such as titanium, aluminum, molybdenum, tungsten, copper, or silver.

(53) The opening **OP2** surrounded by the second layer **32** of the barrier wall **30** has substantially the same size as the opening **OP1** of the insulating layer **12**, but the opening **OP2** may be smaller than the opening **OP1**.

(54) A layer **41** formed of the same material as the organic layer **OR** of the display element **20** and a layer **42** formed of the same material as the upper electrode **E2** are stacked as deposits, on the barrier wall **30**. The layer **41** is in contact with the protective layer **PR**. In addition, the layer **41** is separated from the organic layer **OR** of the display element **20**. The layer **42** is in contact with the layer **41**. In addition, the layer **42** is separated from the upper electrode **E2** of the display element **20**.

(55) The display element **20** and the barrier wall **30** are covered with a sealing film whose illustration in the figure is omitted.

(56) Next, a method of manufacturing the above-described electronic device **100** will be described with reference to FIG. 5 and FIG. 6.

(57) First, as shown in the upper stage in FIG. 5, the pixel circuits **1** and the like shown in FIG. 1 are formed on the base material **10** and then the insulating layer **11** is formed. Then, after the conductive material is formed on the insulating layer **11**, this conductive material is patterned in a predetermined shape, and the lower electrodes **E1** are thereby formed. Then, the insulating layer **12** covering the insulating layer **11** and the lower electrodes **E1** is formed.

(58) Then, as shown in the middle stage, the barrier wall **30** comprising the first layer **31**, the second layer **32**, and the protective layer **PR** is formed on the insulating layer **12**. An example of a method of forming the barrier wall **30** will be described below.

(59) First, the first metal layer is formed to form the first layer **31** on the insulating layer **12**, the insulating layer is formed to form the second layer **32** on the first metal layer, and the second metal layer is formed to form the protective layer **PR** on the insulating layer.

(60) Then, the protective layer PR is formed by patterning (for example, wet etching) the second metal layer into a predetermined shape.

(61) Then, the second layer **32** is formed by etching (for example, dry etching) the insulating layer using the protective layer PR as a mask. In the etching, a resist may be provided to protect the second layer **32**.

(62) Then, the first metal layer is etched (for example, wet etched) using the protective layer PR and the second layer **32** as masks, and the first layer **31** is thereby formed. In the etching, a resist may be provided to protect the protective layer PR and the second layer **32**. The barrier wall **30** is formed through these processes.

(63) The etching of the second metal layer and the insulating layer may be wholly performed, the etching of the first metal layer and the insulating layer may be wholly performed, or the etching of the first metal layer, the insulating layer, and the second metal layer may be wholly performed. In addition, the method of forming the barrier wall **30** is not limited to the method described here.

(64) Then, as shown in the lower stage, the opening OP1 is formed in the insulating layer **12** to expose the lower electrodes E1. At this time, the openings OP1 are formed by dry etching the insulating layer **12** using the barrier wall **30** as a mask.

(65) Thus, since the lower electrodes E1 are exposed from the insulating layer **12** after forming the barrier wall **30**, the lower electrodes E1 are not exposed to an etchant when forming the barrier wall **30**. For example, when the barrier wall **30** is formed after forming the openings OP1, the lower electrodes E1 exposed through the openings OP1 may be damaged by an etchant for forming the first layer **31** of the barrier wall **30** or an etchant for forming the protective layer PR. For this reason, damage to the lower electrodes E1 can be reduced by forming the openings OP1 for exposing the lower electrodes E1 after forming the barrier wall **30** as described above.

(66) In addition, when dry etching the insulating layer **12** using the barrier wall **30** as a mask, the second layer **32** of the barrier wall **30**, which is the insulating layer of the same type as the insulating layer **12**, is protected by the protective layer PR. For this reason, in the dry etching process of the insulating layer **12**, the exposed insulating layer **12** is removed while the second layer **32** covered with the protective layer PR is protected, and the openings OP1 having a desired shape can be formed. The reduction in reliability can be thereby suppressed.

(67) Next, as shown in an upper stage in FIG. 6, the organic layer OR is formed on the lower electrodes E1 by, for example, vacuum deposition, using the barrier wall **30** as a mask. At this time, vapor from an evaporation source passes through the area where the barrier wall **30** does not exist, and reaches upper surfaces of the lower electrodes E1 and the insulating layer **12**. Vapor from the evaporation source does not reach the area in the shadow of the second layer **32** of the barrier wall **30**. The layer **41** of the same material as the organic layer OR is deposited on the protective layer PR.

(68) Then, as shown in the lower stage, the upper electrodes E2 are formed on the organic layer OR by, for example, sputtering, using the barrier wall **30** as a mask. At this time, vapor from the evaporation source passes through the area where the barrier wall **30** does not exist, reaches the upper surface of the organic layer OR and the insulating layer **12** and also reaches the side surfaces S11 and S12 of the first layer **31**. The layer **42** of the same material as the upper electrodes E2 is deposited on the layer **41**.

(69) As described above, the organic layer OR and the upper electrodes E2 of the display element **20** are formed without applying a fine mask. For this reason, the manufacturing costs can be reduced as compared with a case where a fine mask is applied, and processes such as positioning a fine mask are not required, and the organic layer OR and the upper electrodes E2 of the desired shape can easily be formed. In addition, the light emitting areas can be formed in predetermined areas and undesired light emission in the areas overlapping the insulating layer **12** is suppressed, in the display element **20**.

(70) Next, another configuration example of the barrier wall **30** will be described.

(71) FIG. 7 is an enlarged cross-sectional view showing another configuration example of the barrier wall 30. The configuration example shown in FIG. 7 is different from the configuration example shown in FIG. 4 in that the protective layer PR covers the upper surface U21 of the second layer 32 as well as the side surfaces S21 and S22.

(72) An example of a method of forming the barrier wall 30 having such a shape will be described.

(73) First, the first metal layer is formed to form the first layer 31 on the insulating layer 12, and the insulating layer is formed to form the second layer 32 on this first metal layer.

(74) Then, the insulating layer is patterned (for example, dry etched) in a predetermined shape, and the second layer 32 having the upper surface U21 and the side surfaces S21 and S22 is thereby formed.

(75) Then, after forming the second metal layer on the second layer 32, the second metal layer is patterned (for example, wet etched) in a predetermined shape, and the protective layer PR covering the upper surface U21 and the side surfaces S21 and S22 is thereby formed.

(76) Then, the first metal layer is etched (for example, wet etched) using the protective layer PR and the second layer 32 as masks, and the first layer 31 is thereby formed.

(77) After forming the barrier wall 30 in such a shape, the openings OP1 are formed in the insulating layer 12 to expose the lower electrodes E1 as shown in the lower stage in FIG. 5. At this time, the openings OP1 are formed by dry etching the insulating layer 12 using the barrier wall 30 as a mask.

(78) When dry etching the insulating layer 12 using the barrier wall 30 as a mask, the upper surface U21, and the side surfaces S21 and S22 of the second layer 32, which is the insulating layer of the same type as the insulating layer 12, are protected by the protective layer PR. For this reason, the shape of the second layer 32 can be maintained in the dry etching process of the insulating layer 12, and in the process of forming the organic layer OR and the process of forming the upper electrode E2, which are performed subsequently, the second layer 32 can exert the function as the mask for forming the organic layer OR and the upper electrodes E2 having desired shapes.

(79) FIG. 8 is an enlarged cross-sectional view showing the other configuration example of the barrier wall 30.

(80) The configuration example shown in FIG. 8 is different from the configuration example shown in FIG. 4 in that the barrier wall 30 comprises an intermediate layer ML located between the first layer 31 and the second layer 32.

(81) The intermediate layer ML is formed of a material different from that of the second layer 32, for example, the same metal material as that of the protective layer PR. The intermediate layer ML is in contact with the upper surface U11. In addition, the intermediate layer ML extends from the side surface S11 toward the left side of the figure and also extends from the side surface S12 toward the right side of the figure.

(82) The second layer 32 is disposed on the intermediate layer ML and is in contact with the intermediate layer ML. The side surfaces S21 and S22 and the upper surface U21 of the second layer 32 are covered with the protective layer PR. In addition, the protective layer PR is in contact with the intermediate layer ML. In other words, the entire second layer 32 is covered with the protective layer PR and the intermediate layer ML.

(83) For this reason, as shown in the lower stage of FIG. 5, when dry etching the insulating layer 12 using the barrier wall 30 as a mask, the entire second layer 32, which is an insulating layer of the same type as the insulating layer 12, is protected by the protective layer PR and the intermediate layer ML. Therefore, the shape of the second layer 32 can be maintained in the dry etching process of the insulating layer 12. In the process of forming the organic layer OR and the process of forming the upper electrodes E2, the second layer 32 can exert the function as the mask for forming the organic layer OR and the upper electrodes E2 having the desired shapes.

(84) According to the embodiments described above, an electronic device whose reduction in reliability is suppressed and a manufacturing method thereof can be provided.

(85) All of the electronic devices that can be implemented by a person of ordinary skill in the art through arbitrary design changes to the electronic devices described above as embodiments of the present invention come within the scope of the present invention as long as they are in keeping with the spirit of the present invention.

(86) Various types of the modified examples are easily conceivable within the category of the ideas of the present invention by a person of ordinary skill in the art and the modified examples are also considered to fall within the scope of the present invention. For example, additions, deletions or changes in design of the constituent elements or additions, omissions, or changes in condition of the processes arbitrarily conducted by a person of ordinary skill in the art, in the above embodiments, fall within the scope of the present invention as long as they are in keeping with the spirit of the present invention.

(87) In addition, the other advantages of the aspects described in the embodiments, which are obvious from the descriptions of the present specification or which can be arbitrarily conceived by a person of ordinary skill in the art, are considered to be achievable by the present invention as a matter of course.

Claims

1. An electronic device comprising: a substrate; a first insulating layer disposed on the substrate; a lower electrode disposed on the first insulating layer; a second insulating layer disposed on the first insulating layer and including an opening overlapping the lower electrode; an organic layer covering the lower electrode at the opening; an upper electrode covering the organic layer; and a barrier wall disposed on the second insulating layer, wherein the barrier wall comprises: a first layer formed of a metal material and including a first side surface which is in contact with the upper electrode; a second layer disposed above the first layer and extending from the first side surface toward the opening; an intermediate layer located between the first layer and the second layer; and a protective layer formed of a material different from a material of the organic layer and covering at least an upper surface of the second layer; the intermediate layer extends from the first side surface toward the opening, and the protective layer covers a second side surface of the second layer and is in contact with the intermediate layer.
2. The electronic device of claim 1, wherein the second layer is an inorganic insulating layer formed of a same material as a material of the second insulating layer.
3. The electronic device of claim 1, wherein the protective layer is a metal layer formed of a metal material.
4. A method of manufacturing an electronic device, the method comprising: forming a lower electrode on a first insulating layer; forming a second insulating layer covering the first insulating layer and the lower electrode; forming a barrier wall on the second insulating layer; and forming an opening to expose the lower electrode in the second insulating layer, wherein the barrier wall comprises: a first layer formed of a metal material and including a first side surface; a second layer disposed above the first layer and extending from the first side surface toward the opening; an intermediate layer between the first layer and the second layer; and a protective layer covering at least an upper surface of the second layer; the intermediate layer extends from the first side surface toward the opening, the protective layer covers a second side surface of the second layer and is in contact with the intermediate layer, and the opening is formed by dry etching the second insulating layer using the barrier wall as a mask.
5. The method of claim 4, further comprising: forming the opening and then forming an organic layer on the lower electrode in the opening, wherein the organic layer is formed by vacuum deposition using the barrier wall as a mask.
6. The method of claim 5, further comprising: forming the organic layer and then forming an upper

electrode on the organic layer, wherein the upper electrode is formed by sputtering using the barrier wall as a mask.
